

Grado Labs Hemp

When it comes to using unconventional materials, Grado Labs is no stranger to this. Their latest unconventional product, the Grado Labs Hemp headphones, is made using, you guessed it, hemp. Of course, the headphones cost \$420, and the press release was sure to use words like “blunt”, “passed around” and “rolled”. However, Jonathan Grado, VP of marketing, wanted to make it very clear that this is no joke, and that these are still quality headphones and not a marketing gimmick.



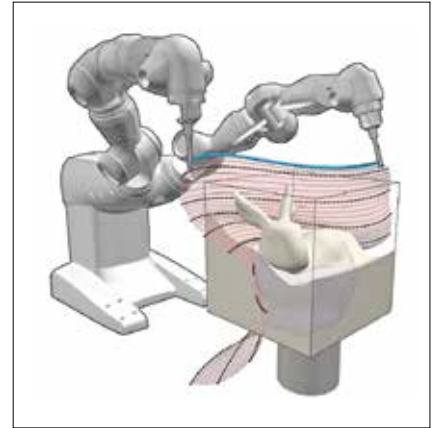
While the idea had been toyed around with in the past, Hemp only became commercially available recently, which is when they actually began experimenting. As it turns out, while hemp is hard to work with, once they actually managed to get it working, it had surprising auditory benefits.

Grado is known for their mid-range-focused sound production, and as it turns out, using hemp seems to have widened their range. This is because compressed hemp creates a damping effect between their fibres, which in turn produces “fuller sound”.

The surprising benefits of using hemp, according to Grado, means that it isn't the last you're going to be seeing of it from them. Check this space out in the future for more hemp headphones!

RoboCut

Researcher's at ETH Zurich's Computational Robotics Lab and the University of Haifa have created a robot that has two hands with wires attached to them. These wires aren't the traditional kind we're accustomed to, they're basically hot wires or rods, which act as carving knives in the hands of RoboCut.



Using these hot wires, researchers used the bot to cut effortlessly through materials like styro-foam and blocks of ice. The advantage of RoboCut comes with flexibility. While traditionally, the hot wire would be straight, RoboCut's wires are flexible and adapt according to the surface it's cutting. In their words, they can “continuously adapt the shape of the tool to match the curvature of the target surface, such that it can be produced more efficiently and with higher quality.”

The team has also come up with several motion-planning algorithms for the robot, which are required for the robot to accurately carve and cut the object it's working on. This required plenty of predicting and simulation of both the required shape of the robot's tools and keeping the shape of the product being carved in mind as well.

Lego Mindstorms Robot Inventor kit

After 7 long years, Lego has finally come out with a new Lego Mindstorms kit. These kits are some of the best toys out there when it comes to teaching kids robotics and programming. The new kit, the Lego Mindstorms Robot Inventor, will let you build up to five different robot models out of 949 pieces. These include robots like a bipedaled, high-fiving, wheeled robot to a four-legged walker robot. Each of the them can be further programmed to perform neat tricks, grab stuff, fire plastic projectiles, avoid obstacles and more.

Included in the kit are stuff like low-profile medium-angular motors, colour and light sensors, distance sensors, and more which work together with the Intelligent Hub block, which is what powers the robots and executes commands. Like with all things Lego, you're not restricted to the five you can build from this kit, you can mix and match and use the tools at hand. The Intelligent Hub is basically the brain of the whole operation, and it comes with six input/output ports for sensors, motors and such. The Hub and all robots can be controlled wirelessly via Bluetooth through the Lego Mindstorms Robot Inventor app for iOS, Android and Windows 10.

